In-locus C-terminal His6 tagging of lasR in PA14 via pEXG2

Primer design • PCR / Assembly protocol • 2026-05-06

1 Konstrukt-Überblick

Zielgen	lasR in PA14 (720 nt, 239 aa)
Strategie	In-locus C-terminal His6 fusion (cassette 30 nt)
Vektor	pEXG2 (5084 bp), HindIII-linearisiert
Methode	In-Fusion 3-Fragment mit Tag-Cassette an Junction
Counterselektion	sacB / GmR (pEXG2)
Finales Plasmid	pEXG2-lasR-His6-C_PA14 • 6806 bp

2 Primer

Primer	Sequenz 5'→ 3' (Tail fett, Body normal)	nt	Tm_{body}	GC
P1_lasR_UP_Fwd	**CATAAATGTAAAGCA**TGTTCTGTGTCGCCGAACTG	35	60.2 °C	55 %
P2_lasR_UP_Rev	**GATGATGATGGTGGCTGCCGCC**GAGAGTAATAAGACCCAAATTAACGG	48	55.9 °C	38 %
P3_lasR_DN_Fwd	**GCCACCATCATCATCACCACTAA**TCTTGCCTCTCAGGTCGGCGAGCTG	48	68.3 °C	64 %
P4_lasR_DN_Rev	**CGACCTGCAGAAGCT**CCCAGCCTTTGCGCTCCTTG	35	63.6 °C	65 %

3 PCR-Bedingungen (B7 High Fidelity)

Tm_{body} Mittel	62.0 °C (Spread 12.41 °C)
Annealing Ta	62 °C (= Tm_{body}; Gradient ± 5 °C empfohlen falls 1. Versuch fehlschlägt)
Elongation	60 s/kb • UP 1266 bp → 76 s • DN 501 bp → 30 s

4 Konstrukt-Zusammenfassung

UP 1266 bp • DN 501 bp • Insert 1752 bp • Final 6806 bp

Tag-Cassette (His6, 30 nt):

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GGC GGC AGC CAC CAT CAT CAT CAC CAC TAA
G G S H H H H H H *
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Off-Target Scan: PASS • HindIII-Sites in finalem Plasmid: 0 (erwartet 0)